Lecture 37-38: Divisibility, Modular Arithmetic, Representations of Integers, Primes

Zhilin Wu (吴志林),
Key Laboratory of System Software,
Institute of Software,
Chinese Academy of Sciences

Outline

- **Introduction to number theory**
- Divisibility
- Modular arithmetic
- Representations of integers
- Greatest common divisors
- Bézout’s Theorem
- Introduction to primes
- Fundamental theorem of arithmetic

Introduction to Number Theory

What is number theory ?

primarily the study of integers and integer-valued functions,

also objects made out of integers (e.g. rational numbers),

or generalizations of integers (e.g. algebraic integers)

Mathematics is the queen of the sciences,

and number theory is the queen of mathematics

Carl Friedrich Gauss (1777–1855), German mathematician

自然科学的皇后是数学，数学的皇冠是数论，

“哥德巴赫猜想”则是皇冠上的明珠

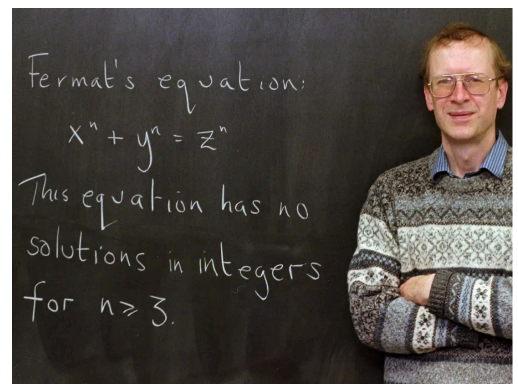
徐迟《哥德巴赫猜想》

Number theory: 数论

Introduction to Number Theory

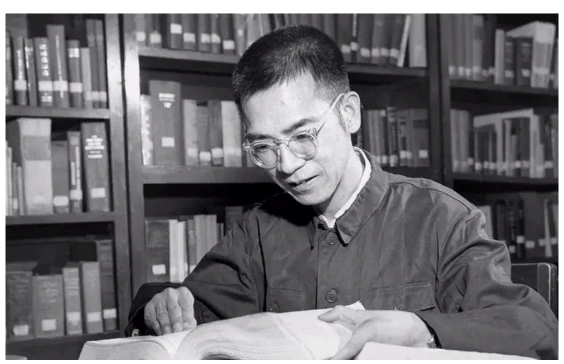
群星璀璨

Pierre de Fermat (1607–1665)
Fermat's Little and Last Theorem

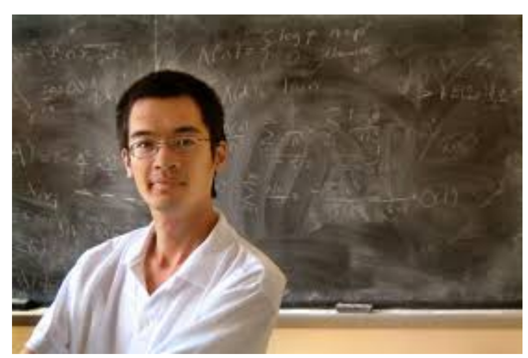


Andrew Wiles (1953 –)
Proved Fermat's Last Theorem

Paul Erdős (1913 – 1996)
First elementary proof for the prime number theorem



陈景润 (1933 – 1996)
Every even number > 2 can be written as the sum of a prime and the product of two primes



Terence Tao 陶哲轩 (1975 –)
There are arbitrarily long arithmetic progressions of prime numbers

Yitang Zhang 张益唐 (1955 –)
There are infinitely many pairs of prime numbers that differ by 70 million or less

divides: 整除 factor: 因子
divisor: 约数 multiple: 倍数

Divisibility

Let a, b $\in \mathbb{Z}$. Then *a divides b*, denoted by $a \mid b$, (a is a *factor/divisor* of b, or b is a *multiple* of a) if $\dfrac{b}{a}$ is an integer.

$a \nmid b$ if a does not divide b.

Proposition. Let a, b, c $\in \mathbb{Z}$ with $a \neq 0$. Then

(i) if $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for all m, $n \in \mathbb{Z}$,

(ii) if $a \mid b$, then $a \mid bc$,

(iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.

Divisibility

Proposition(Division Algorithm). Let $a \in \mathbb{Z}$ and $d \in \mathbb{Z}_+$. Then there are unique q, $r \in \mathbb{Z}$ such that $a = d\,q + r$ and $0 \leq r < d$.

Proof.

We first prove the result for $a \in \mathbb{N}$ by an induction on a.

Induction base $0 \leq a < d$:

$a = d\,0 + a$.

Uniqueness: Suppose $a = dq + r$ for $0 \leq r < d$.

Then $a - r = d\,q$. Therefore, $d \mid (a - r)$.

Because $-d < a - r < d$, we have $a - r = 0$, i.e. $a = r$.

Therefore, $d\,q = 0$, so $q = 0$. This shows $q = 0$, $r = a$ are unique.

Divisibility

Proposition(Division Algorithm). Let $a \in \mathbb{Z}$ and $d \in \mathbb{Z}_+$. Then there are unique q, $r \in \mathbb{Z}$ such that $a = d\,q + r$ and $0 \leq r < d$.

Proof (continued).

Induction step $a \geq d$: Then $0 \leq a - d < a$.

Suppose the result holds for all $a' \in \mathbb{N}$ with $a' < a$.

According to the induction hypothesis, there are unique q', r' such that $a - d = d\,q' + r'$ and $0 \leq r' < d$. Then $a = d\,(q'+1) + r'$.

Uniqueness: Suppose $a = d\,q + r$. Then $d\,(q'+1) + r' = d\,q + r$. Therefore, $d(q'+1-q) = r - r'$. So $d \mid r - r'$. Because $-d < r - r' < d$, we deduce $r - r' = 0$, i.e. $r = r'$. Then $d(q'+1-q) = 0$, thus $q = q'+1$.

Divisibility

Proposition(Division Algorithm). Let $a \in \mathbb{Z}$ and $d \in \mathbb{Z}_+$. Then there are unique $q, r \in \mathbb{Z}$ such that $a = d\,q + r$ and $0 \leq r < d$.

Proof (continued).

We now show the result for $a < 0$.

Since $-a > 0$, there are unique q' and r' such that

$-a = d\,q' + r'$ and $0 \leq r' < d$.

If $r' = 0$, let $q = -q'$ and $r = 0$. Otherwise, let $q = -q'-1$ and $r = d - r'$.

Then $a = d\,q + r$ and $0 \leq r < d$.

Uniqueness: Suppose $a = d\,q'' + r''$ and $0 \leq r'' < d$.

Then $r - r'' = d\,(q'' - q)$. Thus $d \mid (r - r'')$.

If $r = 0$, then $-d < r - r'' \leq 0$, otherwise, $-d < r - r'' < d$.

Therefore, $r - r'' = 0$, i.e. $r'' = r$. Then $d\,(q'' - q) = 0$. Thus $q'' = q$.

divisor: 除数, dividend: 被除数 quotient: 商 remainder: 余数

Divisibility

$$a = d\,q + r \text{ and } 0 \leq r < d$$

divisor: 除数, dividend: 被除数 quotient: 商 remainder: 余数

Divisibility

divisor

$a = d\ q + r$ and $0 \leq r < d$

dividend quotient remainder

divisor: 除数, dividend: 被除数 quotient: 商 remainder: 余数

Divisibility

divisor

a **div** $d = \lfloor a/d \rfloor$

$$a = d\,q + r \text{ and } 0 \leq r < d$$

dividend quotient remainder

divisor: 除数, dividend: 被除数 quotient: 商 remainder: 余数

Divisibility

divisor

a **div** $d = \lfloor a/d \rfloor$

$$a = d\,q + r \text{ and } 0 \le r < d$$

dividend quotient remainder

$$q = a \textbf{ div } d,\ r = a \textbf{ mod } d$$

Example. $5 = 3 \cdot 1 + 2$, $1 = 5$ **div** 3, $2 = 5$ **mod** 3,

$-5 = 3 \cdot (-2) + 1$, $-2 = -5$ **div** 3, $1 = -5$ **mod** 3

Outline

- Introduction to number theory
- Divisibility
- **Modular arithmetic**
- Representations of integers
- Greatest common divisors
- Bézout's Theorem
- Introduction to primes
- Fundamental theorem of arithmetic

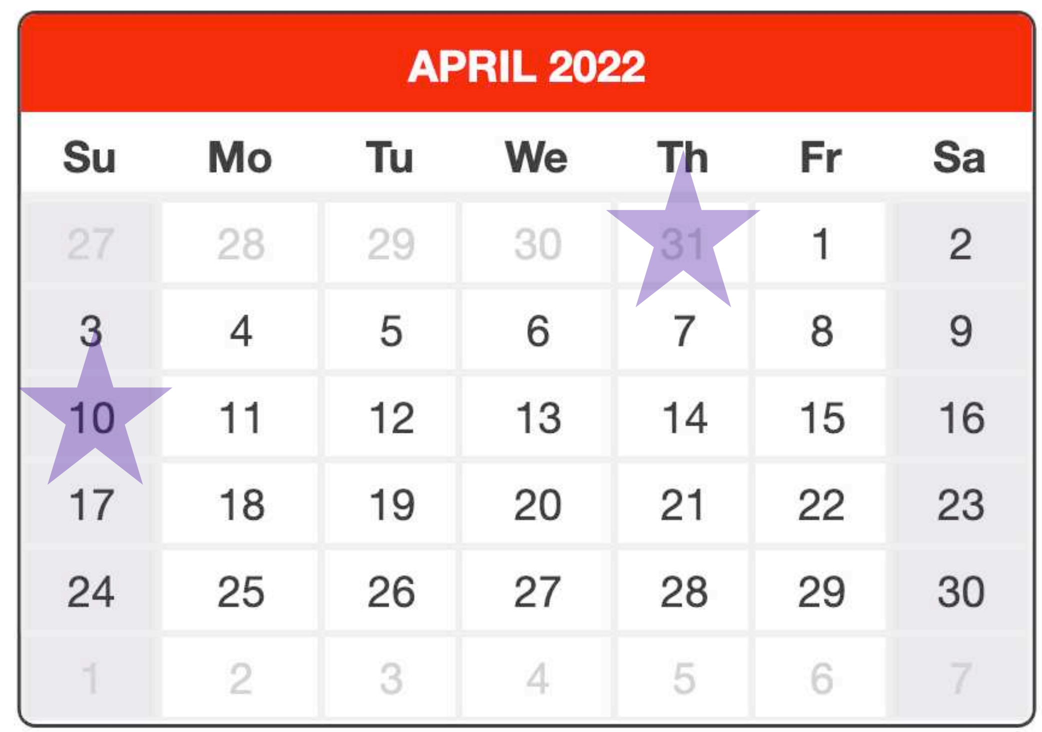
Modular Arithmetic

Why？

In some situations, we only care about remainders.

What weekday is 10 days after 2022/03/31?
(从2022年3月31日开始，之后第十天是星期几？)

0 1 2 3 4 5 6



What weekday is 100 days after 2022/03/31?

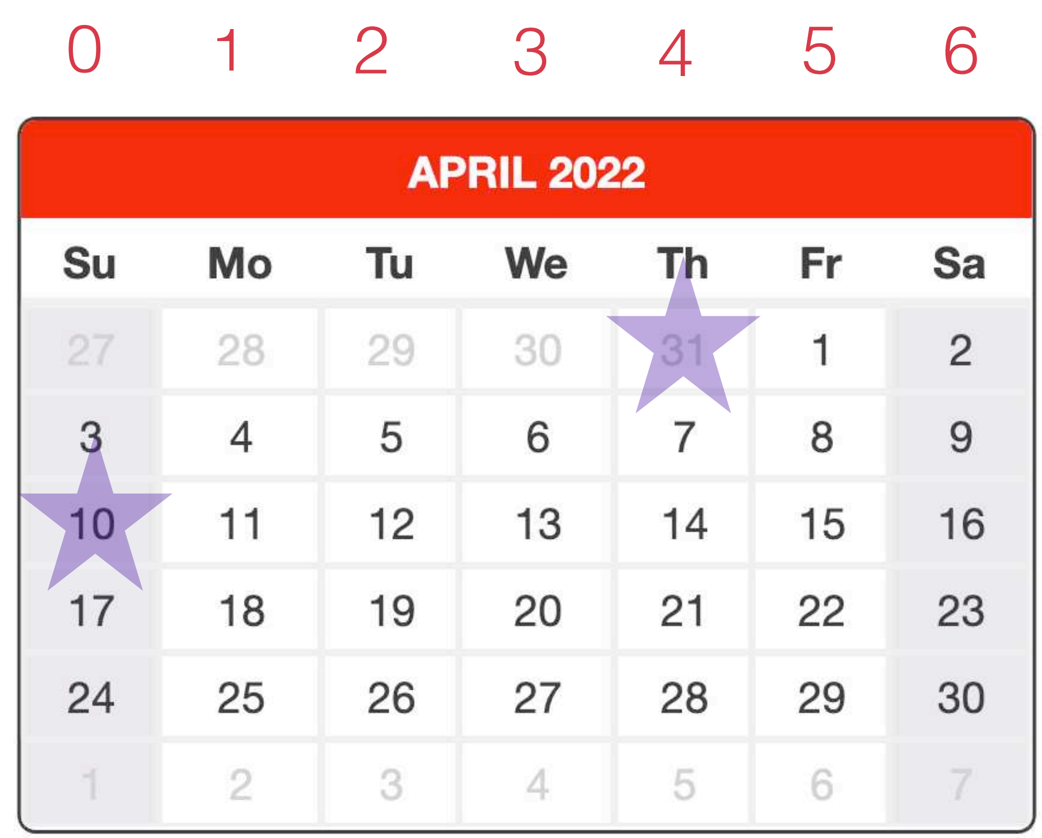
Modular Arithmetic

Why？

In some situations, we only care about remainders.

What weekday is 10 days after 2022/03/31?
(从2022年3月31日开始，之后第十天是星期几？)

$(4+10)$ **mod** $7 = 0$



What weekday is 100 days after 2022/03/31?

Modular Arithmetic

Why?

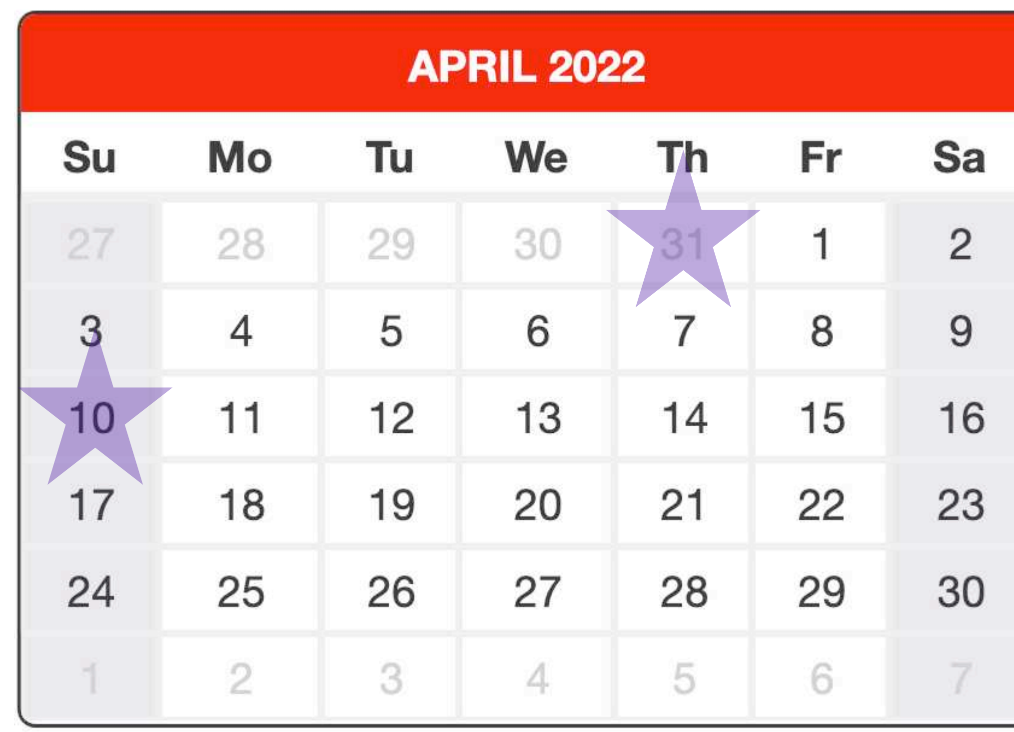
In some situations, we only care about remainders.

What weekday is 10 days after 2022/03/31?

(从2022年3月31日开始，之后第十天是星期几？)

$(4+10)$ **mod** $7 = 0$

0 1 2 3 4 5 6



What weekday is 100 days after 2022/03/31? $(4+100)$ **mod** $7 = 6$

congruence: 同余 modulus: 模数

Modular Arithmetic

Congruences.

Let a, $b \in \mathbb{Z}$ and $m \in \mathbb{Z}_+$.

Then a is *congruent* to b modulo m,

denoted by $a \equiv b \pmod{m}$,

if $m \mid (a - b)$.

$a \equiv b \pmod{m}$ is a *congruence* and m is its *modulus*.

$a \not\equiv b \pmod{m}$ if not $a \equiv b \pmod{m}$.

Proposition. Let a, $b \in \mathbb{Z}$ and $m \in \mathbb{Z}_+$.

Then $a \equiv b \pmod{m}$ iff a **mod** $m = b$ **mod** m.

Modular Arithmetic

Proposition. Let $m \in \mathbb{Z}_+$.

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof.

$a \equiv b \pmod{m} \;\Rightarrow\; a - b = m\,k$ for some $k \in \mathbb{Z}$

$c \equiv d \pmod{m} \;\Rightarrow\; c - d = m\,l$ for some $l \in \mathbb{Z}$

Therefore, $(a + c) - (b + d) = m\,k + m\,l = m(k + l)$.

Thus $a + c \equiv b + d \pmod{m}$.

Modular Arithmetic

Proposition. Let $m \in \mathbb{Z}_+$.

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Proof.

$a \equiv b \pmod{m} \;\Rightarrow\; a - b = m\,k$ for some $k \in \mathbb{Z}$

$c \equiv d \pmod{m} \;\Rightarrow\; c - d = m\,l$ for some $l \in \mathbb{Z}$

Therefore, $(a + c) - (b + d) = m\,k + m\,l = m(k + l)$.

Thus $a + c \equiv b + d \pmod{m}$.

$ac = (m\,k + b)\,(m\,l + d) = m(mkl + dk + bl) + bd$.

Thus $ac - bd = m(mkl + dk + bl)$.

We conclude that $ac \equiv bd \pmod{m}$.

Modular Arithmetic

Proposition. Let $m \in \mathbb{Z}_+$.

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Corollary. Let $m \in \mathbb{Z}_+$. Then

$(a + b)$ **mod** $m = (a$ **mod** $m + b$ **mod** $m)$ **mod** m

(ab) **mod** $m = ((a$ **mod** $m)(b$ **mod** $m))$ **mod** m.

Modular Arithmetic

WARNING!!!

We MUST be careful when working with congruences

Modular Arithmetic

WARNING!!!

We MUST be careful when working with congruences

$ac \equiv bc \pmod{m}$ with $m \geq 2$ does not imply $a \equiv b \pmod{m}$

Example. $2 \cdot 5 \equiv 3 \cdot 5 \pmod{5}$, but $2 \not\equiv 3 \pmod{5}$

Modular Arithmetic

WARNING!!!

We MUST be careful when working with congruences

$ac \equiv bc \pmod{m}$ with $m \geq 2$ does not imply $a \equiv b \pmod{m}$

Example. $2 \cdot 5 \equiv 3 \cdot 5 \pmod{5}$, but $2 \not\equiv 3 \pmod{5}$

$a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ with $m \geq 2$ and c, d positive do not imply $a^c \equiv b^d \pmod{m}$.

Example. $2 \equiv 7 \pmod{5}$, but $2^2 \not\equiv 7^7 \pmod{5}$

$2^2 \equiv 4 \pmod{5}$, $7^7 \equiv 2^7 \pmod{5} \equiv 3 \pmod{5}$

Modular Arithmetic

$\mathbb{Z}_m$: $\{0, 1, \ldots, m\text{-}1\}$.

Arithmetic on $\mathbb{Z}_m$ (*Arithmetic modulo m*):

$a +_m b = (a + b)$ **mod** m,

$a \cdot_m b = (a \cdot b)$ **mod** m.

Example.

$4 +_7 5 = 9$ **mod** $7 = 2$

$4 \cdot_7 5 = 20$ **mod** $7 = 6$

Modular Arithmetic

$\mathbb{Z}_m$: $\{0, 1, \ldots, m\text{-}1\}$.

Arithmetic on $\mathbb{Z}_m$ (*Arithmetic modulo m*):

$a +_m b = (a + b)$ **mod** m,

$a \cdot_m b = (a \cdot b)$ **mod** m.

Closure: $\forall\, a, b \in \mathbb{Z}_m\ (a +_m b \in \mathbb{Z}_m$ *and* $a \cdot_m b \in \mathbb{Z}_m)$.

Associativity: $(a +_m b) +_m c = a +_m (b +_m c)$, $(a \cdot_m b) \cdot_m c = a \cdot_m (b \cdot_m c)$.

Commutativity: $a +_m b = b +_m a$, $a \cdot_m b = b \cdot_m a$.

Identity elements: $a +_m 0 = a$, $a \cdot_m 1 = a$.

Additive inverses: If $a \neq 0$, then $a +_m (m - a) = 0$. Moreover, $0 +_m 0 = 0$.

Distributivity: $a \cdot_m (b +_m c) = (a \cdot_m b) +_m (a \cdot_m c)$

Outline

- Introduction to number theory
- Divisibility
- Modular arithmetic
- **Representations of integers**
- Greatest common divisors
- Bézout's Theorem
- Introduction to primes
- Fundamental theorem of arithmetic

Representations of Integers

Integers can be represented
using any base greater than 1.
We commonly use the following representations,

decimal (base 10), 十进制
binary (base 2), 二进制
octal (base 8), 八进制
hexadecimal (base 16). 十六进制

Representations of Integers

Theorem. Let b be an integer greater than 1. Then if $n \in \mathbb{Z}_+$, it can be represented uniquely in the form

$$n = a_k b^k + a_{k-1} b^{k-1} + \ldots + a_1 b + a_0, \text{ where}$$

$k \geq 0$, $a_0, \ldots, a_k \in \mathbb{N}$, $0 \leq a_0, \ldots, a_k < b$, and $a_k \neq 0$.

Representations of Integers

Theorem. Let b be an integer greater than 1. Then if $n \in \mathbb{Z}_+$, it can be represented uniquely in the form

$$n = a_k b^k + a_{k-1} b^{k-1} + \ldots + a_1 b + a_0, \text{ where}$$

$k \geq 0$, $a_0, \ldots, a_k \in \mathbb{N}$, $0 \leq a_0, \ldots, a_k < b$, and $a_k \neq 0$.

Base-b *expansion* of n: $(a_k a_{k-1} \ldots a_1 a_0)_b$

Representations of Integers

Lemma. Let $b \in \mathbb{Z}_+ \setminus \{1\}$, $k \geq 0$, $a_0, \ldots, a_k \in \mathbb{N}$, and $0 \leq a_0, \ldots, a_k < b$. Then $a_k b^k + a_{k-1} b^{k-1} + \ldots + a_1 b + a_0 < b^{k+1}$.

Representations of Integers

Proof of the theorem. By an induction on n.

Representations of Integers

Representations of Integers

Representations of Integers

Decimal	0	1	2	3	4	5	6	7
Hexadecimal	0	1	2	3	4	5	6	7
Octal	0	1	2	3	4	5	6	7
Binary	0	1	10	11	100	101	110	111

Decimal	8	9	10	11	12	13	14	15
Hexadecimal	8	9	A	B	C	D	E	F
Octal	10	11	12	13	14	15	16	17
Binary	1000	1001	1010	1011	1100	1101	1110	1111

Representations of Integers

Examples. Binary, octal, and hexadecimal representations.

$(101011111)_2$

$= 1 \cdot 2^8 + 0 \cdot 2^7 + 1 \cdot 2^6 + 0 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2 + 1$

$= 256 + 64 + 16 + 8 + 4 + 2 + 1 = 351.$

$(7016)_8$

$= 7 \cdot 8^3 + 0 \cdot 8^2 + 1 \cdot 8 + 6 = 3598$

$(2AE0B)_{16}$

$= 2 \cdot 16^4 + 10 \cdot 16^3 + 14 \cdot 16^2 + 0 \cdot 16 + 11 = 175{,}627.$

Base Conversion

Convert a positive integer n into its base-b expansion.

An illustration of the idea

Suppose the binary expansion of n is $(a_k a_{k-1} \ldots a_1 a_0)_2$, i.e.

$n = a_k 2^k + a_{k-1} 2^{k-1} + \ldots + a_1 2 + a_0$.

Then $n = 2\, n_1 + a_0$ with $n_1 = a_k 2^{k-1} + a_{k-1} 2^{k-2} + \ldots + a_1$.

Therefore, $a_0 = n$ **mod** 2 and $n_1 = n$ **div** 2.

Base Conversion

Convert a positive integer n into its base-b expansion.

An illustration of the idea

Suppose the binary expansion of n is $(a_k a_{k-1} \ldots a_1 a_0)_2$, i.e.

$n = a_k 2^k + a_{k-1} 2^{k-1} + \ldots + a_1 2 + a_0$.

Then $n = 2 n_1 + a_0$ with $n_1 = a_k 2^{k-1} + a_{k-1} 2^{k-2} + \ldots + a_1$.

Therefore, $a_0 = n$ **mod** 2 and $n_1 = n$ **div** 2.

Repeat the arguments, we have

$a_1 = n_1$ **mod** 2 and $n_2 = n_1$ **div** 2 with $n_2 = a_k 2^{k-2} + a_{k-1} 2^{k-3} + \ldots + a_2$,

Base Conversion

Convert a positive integer n into its base-b expansion.

An illustration of the idea

Suppose the binary expansion of n is $(a_k a_{k-1} \ldots a_1 a_0)_2$, i.e.

$$n = a_k 2^k + a_{k-1} 2^{k-1} + \ldots + a_1 2 + a_0.$$

Then $n = 2\, n_1 + a_0$ with $n_1 = a_k 2^{k-1} + a_{k-1} 2^{k-2} + \ldots + a_1$.

Therefore, $a_0 = n$ **mod** 2 and $n_1 = n$ **div** 2.

Repeat the arguments, we have

$a_1 = n_1$ **mod** 2 and $n_2 = n_1$ **div** 2 with $n_2 = a_k 2^{k-2} + a_{k-1} 2^{k-3} + \ldots + a_2$,

$\ldots$,

$a_k = n_k$ **mod** 2 and $n_{k+1} = n_k$ **div** $2 = 0$.

Base Conversion

Convert a positive integer *n* into its base-*b* expansion.

procedure *base b expansion*(n, b: positive integers with $b > 1$)
$q := n$
$k := 0$
while $q \neq 0$
    $a_k := q$ **mod** b
    $q := q$ **div** b
    $k := k + 1$
return $(a_{k-1}, \ldots, a_1, a_0)$ $\{(a_{k-1} \ldots a_1 a_0)_b$ is the base b expansion of $n\}$

Base Conversion

$$(\cdots)_2 \Rightarrow (\cdots)_8$$

$$(1\ 1\ 0\ 1\ 0\ 1\ 1\ 1\ 1\ 1)_2$$

Base Conversion

$$(\cdots)_2 \Rightarrow (\cdots)_8$$

$$(\underbrace{1}\ \underbrace{1\ 0\ 1}\ \underbrace{0\ 1\ 1}\ \underbrace{1\ 1\ 1})_2$$

Base Conversion

$$(\cdots)_2 \Rightarrow (\cdots)_8$$

$$(\underbrace{1}\ \underbrace{1\ 0\ 1}\ \underbrace{0\ 1\ 1}\ \underbrace{1\ 1\ 1})_2$$

$$(1\quad 5\quad 3\quad 7)_8$$

Base Conversion

$$(\cdots)_8 \Rightarrow (\cdots)_2$$

$$(1 \quad 5 \qquad\quad 3 \qquad\quad 7)_8$$

Base Conversion

$$(\cdots)_8 \Rightarrow (\cdots)_2$$

$$(1 \quad 5 \quad 3 \quad 7)_8$$

$$(\overbrace{1}\ \overbrace{1\,0\,1}\ \overbrace{0\,1\,1}\ \overbrace{1\,1\,1})_2$$

Base Conversion

$$(\cdots)_2 \Rightarrow (\cdots)_{16}$$

$$(1\ 1\ 0\ 1\ 0\ 1\ 1\ 1\ 1\ 1)_2$$

Base Conversion

$$(\cdots)_2 \Rightarrow (\cdots)_{16}$$

$$(\underbrace{1\ 1}\ \underbrace{0\ 1\ 0\ 1}\ \underbrace{1\ 1\ 1\ 1})_2$$

Base Conversion

$$(\cdots)_2 \Rightarrow (\cdots)_{16}$$

$$(\underbrace{1\ 1}\ \underbrace{0\ 1\ 0\ 1}\ \underbrace{1\ 1\ 1\ 1})_2$$

$$(3 \quad 5 \quad F)_{16}$$

Base Conversion

$$(\cdots)_{16} \Rightarrow (\cdots)_2$$

$$(3 \qquad 5 \qquad F)_{16}$$

Base Conversion

$$(\cdots)_{16} \Rightarrow (\cdots)_2$$

$$(3 \qquad 5 \qquad F)_{16}$$

$$(\overbrace{1\ 1}\ \overbrace{0\ 1\ 0\ 1}\ \overbrace{1\ 1\ 1\ 1})_2$$

Outline

- Introduction to number theory
- Divisibility
- Modular arithmetic
- Representations of integers
- **Greatest common divisors**
- Bézout's Theorem
- Introduction to primes
- Fundamental theorem of arithmetic

Greatest Common Divisors

The *greatest common divisor* of two integers a and b
such that $a \neq 0$ or $b \neq 0$,
denoted by $gcd(a, b)$,
is the largest positive integer d such that $d \mid a$ and $d \mid b$.

Examples.

$gcd(4, 6) = 2$, $gcd(-4, -6) = 2$, $gcd(5, 10) = 5$,

$gcd(5, 7) = 1$, $gcd(0, 5) = 5$, $gcd(0, -5) = 5$.

Greatest Common Divisors

The *greatest common divisor* of two integers a and b
such that $a \neq 0$ or $b \neq 0$,
denoted by $gcd(a, b)$,
is the largest positive integer d such that $d \mid a$ and $d \mid b$.

Examples.

$gcd(4, 6) = 2$, $gcd(-4, -6) = 2$, $gcd(5, 10) = 5$,

$gcd(5, 7) = 1$, $gcd(0, 5) = 5$, $gcd(0, -5) = 5$.

Proposition. $gcd(a, b) = gcd(-a, b) = gcd(a, -b) = gcd(-a, -b)$.

Greatest Common Divisors

The *greatest common divisor* of two integers a and b
such that $a \neq 0$ or $b \neq 0$,
denoted by $gcd(a, b)$,
is the largest positive integer d such that $d \mid a$ and $d \mid b$.

Examples.

$gcd(4, 6) = 2$, $gcd(-4, -6) = 2$, $gcd(5, 10) = 5$,

$gcd(5, 7) = 1$, $gcd(0, 5) = 5$, $gcd(0, -5) = 5$.

Proposition. $gcd(a, b) = gcd(-a, b) = gcd(a, -b) = gcd(-a, -b)$.

Why not $gcd(0, 0)$?

Greatest Common Divisors

The *greatest common divisor* of two integers a and b
such that $a \neq 0$ or $b \neq 0$,
denoted by $gcd(a, b)$,
is the largest positive integer d such that $d \mid a$ and $d \mid b$.

Examples.

$gcd(4, 6) = 2$, $gcd(-4, -6) = 2$, $gcd(5, 10) = 5$,
$gcd(5, 7) = 1$, $gcd(0, 5) = 5$, $gcd(0, -5) = 5$.

Proposition. $gcd(a, b) = gcd(-a, b) = gcd(a, -b) = gcd(-a, -b)$.

Why not $gcd(0, 0)$? Any number n is a divisor of 0

relatively prime: 互素

Greatest Common Divisors

Two integers *a*, *b* are *relatively prime* if $gcd(a, b)$ is defined and $gcd(a, b) = 1$.

Examples.

5 and 7 are relatively prime,

4 and 6 (resp. 0 and 5) are not relatively prime.

relatively prime: 互素

Greatest Common Divisors

Two integers a, b are *relatively prime* if $gcd(a, b)$ is defined and $gcd(a, b) = 1$.

Examples.

5 and 7 are relatively prime,

4 and 6 (resp. 0 and 5) are not relatively prime.

The integers $a_1, \ldots, a_n$ are *pairwise relatively prime* if a_i and a_j are relatively prime for every $1 \leq i < j \leq n$.

Greatest Common Divisors

Two integers a, b are *relatively prime* if $gcd(a, b)$ is defined and $gcd(a, b) = 1$.

Examples.

5 and 7 are relatively prime,

4 and 6 (resp. 0 and 5) are not relatively prime.

The integers $a_1, \ldots, a_n$ are *pairwise relatively prime* if a_i and a_j are relatively prime for every $1 \leq i < j \leq n$.

Examples.

5, 7, 9 are pairwise relatively prime,

5, 7, 10 are not pairwise relatively prime.

Bézout's Theorem

Bézout's Theorem. If $a, b \in \mathbb{Z}$ with $a \neq 0$ or $b \neq 0$, then there exist $s, t \in \mathbb{Z}$ such that $gcd(a, b) = sa + tb$.

Examples.

$gcd(5, 7) = 1 = (-4) \cdot 5 + 3 \cdot 7$.

$gcd(4, 6) = 2 = (-4) \cdot 4 + 3 \cdot 6$.

Corollary. If a and b are relatively prime integers, then there exist integers s and t such that $sa + tb = 1$.

Étienne Bézout (1730–1783): French mathematician.

Bézout's Theorem

Bézout's Theorem. If $a, b \in \mathbb{Z}$ with $a \neq 0$ or $b \neq 0$, then there exist $s, t \in \mathbb{Z}$ such that $gcd(a, b) = sa + tb$.

Bézout's identity

Bézout's coefficients

Examples.

$gcd(5, 7) = 1 = (-4) \cdot 5 + 3 \cdot 7.$

$gcd(4, 6) = 2 = (-4) \cdot 4 + 3 \cdot 6.$

Corollary. If a and b are relatively prime integers, then there exist integers s and t such that $sa + tb = 1$.

Étienne Bézout (1730–1783): French mathematician.

relatively prime: 互素

Bézout's Theorem

Proof.

Outline

- Introduction to number theory
- Divisibility
- Modular arithmetic
- Representations of integers
- Greatest common divisors
- Bézout’s Theorem
- **Introduction to primes**
- Fundamental theorem of arithmetic

Prime: 素数 Composite: 合数

Twin primes: 孪生素数
Public-key cryptography: 公钥密码体系

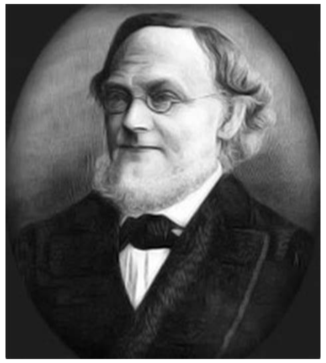
Introduction to Primes

What is a *prime* ?

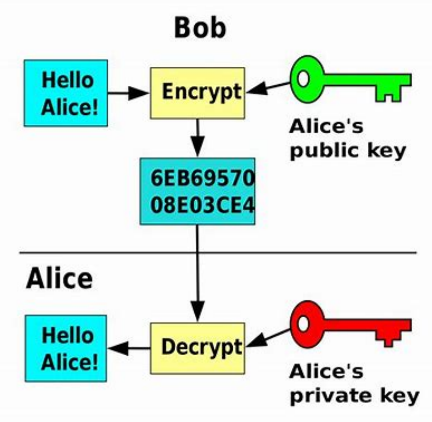
A positive integer $p > 1$

that has only two positive factors 1 and p.

If p is not a prime, then it is called a *composite*.



- Central in number theory: Many famous theorems and conjectures in number theory are about primes.
  - Fundamental theorem of arithmetic, Prime number theorem,
  - Goldbach's conjecture: the “1+1” problem, every even integer > 2 can be expressed as the sum of two primes,
  - Twin prime conjecture: there are infinitely many pairs of twin primes (a pair of primes differing by two).
- Important application in public-key cryptography, relying on the difficulty of factorizing large numbers into their prime factors



Mersenne prime: 梅森素数

Introduction to Primes

What is a *prime*?

A positive integer $p > 1$

that has only two positive factors 1 and p.

If p is not a prime, then it is called a *composite*.

- There are infinitely many primes:

  First proved by Euclid around 300 BC.

- Primality test: In polynomial time, Manindra Agrawal, Neeraj Kayal, and Nitin Saxena, 2002.

- The largest known prime number: $2^{82,589,933} - 1$

  24,862,048 decimal digits, 6,629 pages to display it (50 lines/page, 75 digits/line),

  Patrick Laroche, December 2018,

  Great Internet Mersenne Prime Search (GIMPS).

Fundamental Theorem of Arithmetic

THE FUNDAMENTAL THEOREM OF ARITHMETIC

Let $n \in \mathbb{Z}_+$ with $n > 1$.

Then n can be written uniquely into the form

$$p_1^{i_1} \cdot \ldots \cdot p_k^{i_k}$$

such that $p_1, \ldots, p_k$ are primes,

$p_1 < \ldots < p_k$, and $i_1, \ldots, i_k > 0$.

Examples. $100 = 2^2 \cdot 5^2$ and $999 = 3^3 \cdot 37^1$.

Fundamental theorem of arithmetic: 算术基本定理
Prime factorization: 素因子分解

Fundamental Theorem of Arithmetic

THE FUNDAMENTAL THEOREM OF ARITHMETIC

Let $n \in \mathbb{Z}_+$ with $n > 1$.

Then n can be written uniquely into the form

$$p_1^{i_1} \cdot \ldots \cdot p_k^{i_k} \longleftarrow \text{Prime factorization}$$

such that $p_1, \ldots, p_k$ are primes,

$p_1 < \ldots < p_k$, and $i_1, \ldots, i_k > 0$.

Examples. $100 = 2^2 \cdot 5^2$ and $999 = 3^3 \cdot 37^1$.

Fundamental Theorem of Arithmetic

Lemma 1.
Suppose $a, b, c \in \mathbb{Z}$, $a \mid (bc)$ and $gcd(a, b) = 1$. Then $a \mid c$.

Fundamental Theorem of Arithmetic

Corollary. Let $m \in \mathbb{Z}_+$ and a, b, $c \in \mathbb{Z}$.

If $ac \equiv bc \pmod{m}$ and $gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Proof.

From $ac \equiv bc \pmod{m}$, we know that $m \mid (ac - bc) = (a-b)c$.

Because $gcd(c, m) = 1$, from **Lemma 1**, we deduce $m \mid (a - b)$.

Therefore, $a \equiv b \pmod{m}$.

Fundamental Theorem of Arithmetic

Lemma 2.
Suppose p is a prime and $p|(a_1 \ldots a_n)$. Then $p \mid a_i$ for some i.

Proof. Induction on n.

Induction base $n = 1$: Trivial.

Induction step $n > 1$: Suppose the lemma holds for every $n' < n$.

If $p|a_1$, then we are done.

Otherwise, $\gcd(p, a_1)=1$.

Since $p|a_1(a_2 \ldots a_n)$, from **Lemma 1**, we have $p \mid a_2 \ldots a_n$.

According to the induction hypothesis, $p \mid a_i$ for some i: $2 \leq i \leq n$.

Fundamental Theorem of Arithmetic

Proof of the theorem. Induction on n.

Induction base: $n = 2$. Then $n = 2^1$ since 2 is a prime.

Induction step:

Suppose $n > 2$ and the theorem holds for all $n' < n$.

Case n is a prime

Fundamental Theorem of Arithmetic

Proof of the theorem (continued). Induction on n.

Induction step:

Case n is not a prime:

Fundamental Theorem of Arithmetic

Proof of the theorem (continued).

Case n is not a prime:

Uniqueness:

What is number theory ?
primarily the study of integers and integer-valued functions,
also objects made out of integers (e.g. rational numbers),
or generalizations of integers (e.g. algebraic integers)

Let a, $b \in \mathbb{Z}$. Then a *divides* b, denoted by $a \mid b$,
(a is a *factor*/*divisor* of b, or b is a *multiple* of a)
if $\dfrac{b}{a}$ is an integer.
$a \nmid b$ if a does not divide b.

Recap

Proposition(Division Algorithm). Let $a \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$. Then there are unique q, $r \in \mathbb{Z}$ such that $a = d\,q + r$ and $0 \leq r < d$.

divisor

$$a = d\,q + r \text{ and } 0 \leq r < d$$

dividend quotient remainder

$q = a$ **div** d, $r = a$ **mod** d

- Introduction to number theory
- Divisibility
- Modular arithmetic
- Representations of integers

Congruences.
Let a, $b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$.
Then a is *congruent* to b modulo m,
denoted by $a \equiv b \pmod{m}$,
if $m \mid (a - b)$.
$a \equiv b \pmod{m}$ is a *congruence* and m is its *modulus*.
$a \not\equiv b \pmod{m}$ if not $a \equiv b \pmod{m}$.

$\mathbb{Z}_m$: $\{0, 1, \ldots, m\text{-}1\}$.
Arithmetic on $\mathbb{Z}_m$ (*Arithmetic modulo m*):
$a +_m b = (a + b)$ **mod** m,
$a \cdot_m b = (a \cdot b)$ **mod** m.

Associativity, commutativity, distributivity, …

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then
$a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Decimal	0	1	2	3	4	5	6	7
Hexadecimal	0	1	2	3	4	5	6	7
Octal	0	1	2	3	4	5	6	7
Binary	0	1	10	11	100	101	110	111

Decimal	8	9	10	11	12	13	14	15
Hexadecimal	8	9	A	B	C	D	E	F
Octal	10	11	12	13	14	15	16	17
Binary	1000	1001	1010	1011	1100	1101	1110	1111

Theorem. Let b be an integer greater than 1. Then if $n \in \mathbb{Z}^+$, it can be represented uniquely in the form
$n = a_k b^k + a_{k-1} b^{k-1} + \ldots + a_1 b + a_0$, where
$k \geq 0$, $a_0, \ldots, a_k \in \mathbb{N}$, $0 \leq a_0, \ldots, a_k < b$, and $a_k \neq 0$.

$$(a_k a_{k-1} \ldots a_1 a_0)_b$$

procedure *base b expansion*(n, b: positive integers with $b > 1$)
$q := n$
$k := 0$
while $q \neq 0$
  $a_k := q$ **mod** b
  $q := q$ **div** b
  $k := k + 1$
return $(a_{k-1}, \ldots, a_1, a_0)$ $\{(a_{k-1} \ldots a_1 a_0)_b$ is the base b expansion of $n\}$

Recap

The *greatest common divisor* of two integers a and b
such that $a \neq 0$ or $b \neq 0$,
denoted by $gcd(a, b)$,
is the largest positive integer d such that $d \mid a$ and $d \mid b$.

Two integers a, b are *relatively prime* if
$gcd(a, b)$ is defined and $gcd(a, b) = 1$.

- Greatest common divisors
- Bézout's Theorem
- Fundamental theorem of arithmetic

Bézout's Theorem. If $a, b \in \mathbb{Z}$ with $a \neq 0$ or $b \neq 0$,
then there exist $s, t \in \mathbb{Z}$ such that $gcd(a, b) = sa + tb$.

THE FUNDAMENTAL THEOREM OF ARITHMETIC
Let $n \in \mathbb{Z}^+$ with $n > 1$.
Then n can be written uniquely into the form

$$p_1^{i_1} \cdot \ldots \cdot p_k^{i_k}$$

such that $p_1, \ldots, p_k$ are primes,
$p_1 < \ldots < p_k$, and $i_1, \ldots, i_k > 0$.

Lemma 1.
Suppose $a, b, c \in \mathbb{Z}$, $a \mid (bc)$ and $gcd(a, b) = 1$. Then $a \mid c$.

Lemma 2.
Suppose p is a prime and $p \mid a_1 \ldots a_n$. Then $a \mid a_i$ for some i.

Corollary. Let $m \in \mathbb{Z}^+$ and $a, b, c \in \mathbb{Z}$.
If $ac \equiv bc \pmod{m}$ and $gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Homework

1. Section 4.1.
   - Page 259, Exercise 46, 48.
2. Section 4.2.
   - Page 269, Exercise 18.
2. Section 4.3.
   - Page 289, Exercise 10.

作业和答疑

1. 交作业时间：奇数周周一上课前（两周交一次）
2. 交作业方式（二选一）：
   - 通过https://mooc.ucas.edu.cn/portal课程网站
   - 上课现场提交
3. 答疑：助教崔乐天、刘易铖

Next Lecture

Primes, Euclidean algorithm,
Congruences